Experiment 14: Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

Aim

To learn the basics of Deluge, the scripting language of Zoho Creator, and to write simple programs and custom functions.

Objective

- Understand the syntax and data types of Deluge.
- Write programs using variables, conditions and loops.
- Create and call a custom function.
- Use Deluge to read and update form records.

Introduction

Deluge (Data Enriched Language for the Universal Grid Environment) is an online scripting language used in Zoho products. It is used in workflows, buttons and functions to add logic that drag-and-drop settings cannot give. Every statement ends with a semicolon, and comments start with `//`.

Data Type	Example
Text	name = "Ravi";
Number	count = 5;
Decimal	amount = 1250.50;
Boolean	isActive = true;
Date	d = '15-Oct-2026';
List	items = {"A", "B"};
Map	m = Map(); m.put("key", "value");

Where to Write Deluge

Open the application, go to **Workflows > Functions > Create New Function** for custom functions, or use the script builder in form workflows. The **Execute Script** option shows output of `info` statements.

Program 1: Hello and Variables

```
name = "Ravi";
marks = 78;
info "Hello " + name;
info "Marks: " + marks;
```

Program 2: Conditions

```
marks = 78;
if(marks >= 90)
{
    info "Grade A";
}
else if(marks >= 60)
{
    info "Grade B";
}
else
{
    info "Grade C";
}
```

Program 3: Loop over a List

```
depts = {"HR", "Finance", "IT", "Sales"};
for each d in depts
{
    info "Department: " + d;
}
```

Program 4: Custom Function

Create a function in the Functions section that returns the number of leave days between two dates.

```
int LeaveUtil.getLeaveDays(date fromDate, date toDate)
{
    days = fromDate.daysBetween(toDate) + 1;
    return days;
}
```

Calling the function from a workflow:

```
total = thisapp.LeaveUtil.getLeaveDays(input.From_Date, input.To_Date);
info "Total leave days: " + total;
```

Program 5: Work with Form Records

Fetch all pending requests and count them:

```
pending = Leave_Request[Status == "Pending"];
info "Pending requests: " + pending.count();
for each r in pending
{
    info r.Employee_Name + " - " + r.Leave_Type;
}
```

Syntax such as the function namespace and the exact date methods can differ slightly between versions. If the editor shows an error, check the suggestion shown by the Deluge editor or the Zoho Creator help documentation.

Attach screenshot here - script in editor

Attach screenshot here - output of info statements

Results

Basic Deluge programs using variables, conditions, loops, a custom function and form record access were written and executed successfully.

Conclusion

Deluge extends no-code applications with real programming logic while keeping the syntax simple and readable. Custom functions make the code reusable across forms and workflows.